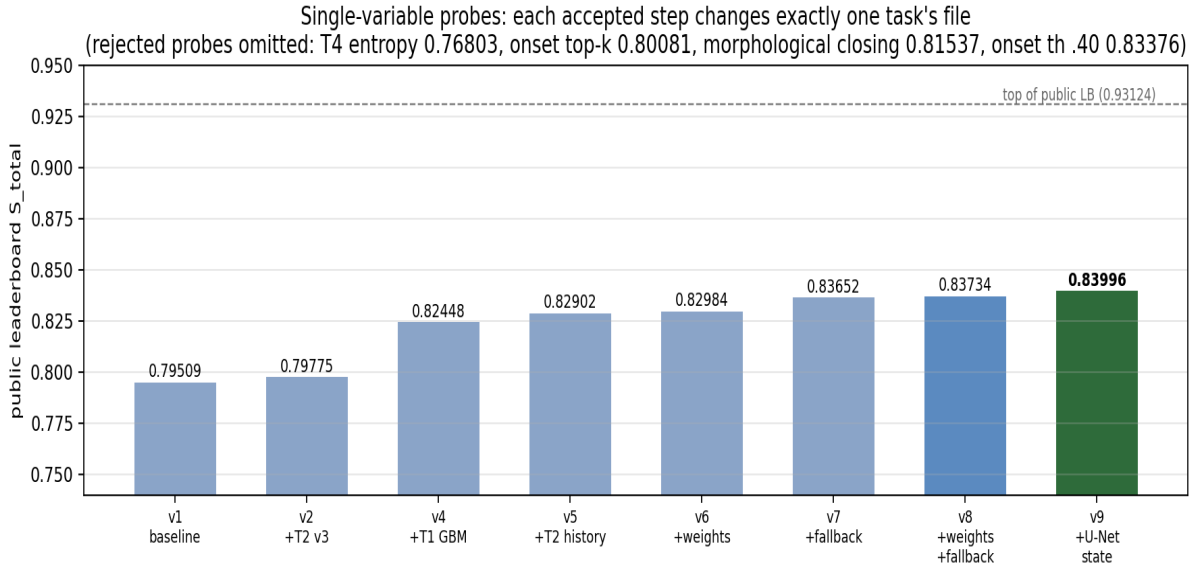
Reconstructing a freeway state field under masking: four coupled tasks on ten corridors

2026 IEEE Big Data Cup — Challenge 06, Track 2 (TrafficFlowBench, Kaggle)
Team CAMPUS SUNNY



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0. Summary

The benchmark asks four coupled questions about ten directional freeway corridors (5-minute resolution, one mainline detector per link, 44–352 links per corridor): reconstruct masked speed/flow (Task 1), forecast queue propagation over the next 30 minutes (Task 2), obey freeway physics (Task 3, scored on the Task 1 answer), and invert link counts into path flows (Task 4). The score is

`S_total = 0.35·S_state + 0.30·S_queue + 0.15·S_physics + 0.20·S_ODME`.

We proceeded task by task with a train-only evidence base: the `train` split is the only one with unmasked observations, so every model is fitted on `train` and validated by held-out train months, never on the scored months. Each Kaggle submission then changed exactly one task's file, turning the public leaderboard into a controlled attribution experiment for that task.

Final submitted configuration (public LB **0.83996** as of 2026-11-04; the last submission may change after further refinement probes):

task	model	local evidence
1 state	0.6 x LightGBM residual model + 0.4 x compact U-Net on daily corridor images (U-Net refit on all 273 train days)	holdout S_state 0.9367 (GBM) → 0.9408 (blend) on four panels; LB +0.0026
2 queue	LightGBM v3 (onset last-step probability, ongoing per link × step) blended with a long-history model (onset w 0.3, ongoing w 0.4/th 0.20); onset threshold 0.30 **plus a top-1 fallback for windows that would predict nothing**	honest Feb pseudo-CV 0.8195 and released-June proxy 0.8292 after blending; the fallback is justified by the scoring spec (every onset window contains a queue) and worth +0.0075 on the LB
3 physics	no file — scored on the Task 1 rows	calibrated proxy S_physics ≈ 0.85 for the LightGBM state
4 ODME	non-negative least squares with prior regularisation	local S_link 0.99993

Public leaderboard progression (each probe changes one task, everything else held fixed): 0.79509 (all baselines) → 0.79775 (Task 2 v3) → 0.82448 (Task 1 LightGBM) → 0.82902 (Task 2 long-history blend) → 0.82984 (blend weights) → **0.83652** (onset fallback) → **0.83734** (weights + fallback) → **0.83996** (U-Net state blend). Rejected probes: Task 4 entropy reallocation (0.76803), Task-2 ongoing threshold 0.40 (0.82197), Task-2 onset top-3 (0.80081), Task-2 morphological closing (0.81537), onset threshold 0.25 (0.82852), ongoing threshold 0.15 (0.82431, neutral).

1. Data and constraints that shaped the method

- Three splits: `train` = 2030-06-01…2031-02-28 (273 days, unmasked layer

released), `validation` = 2031-03 (public LB), `private` = 2031-04 (final ranking). The two scored months were generated independently (5 and 7 incidents respectively), so a method tuned on the public month does not transfer; we tuned nothing on the public month.

- Only cells with `pct_observed >= 75` and non-null scored channels are scored

(≈76% of cells). Ineligible cells are still released and are part of the problem.

- Task 1 masks: per calendar day, one of three regimes removes 20/30/50% of the

eligible cells (R1/R2/R3); the required row set is enumerated by the released templates, not by looking for nulls (the Task-2 window gaps are blank too).

- A ramp observation that is unavailable is unavailable evidence, never a

measured zero; the physics evaluator excludes such ramp transitions.

- Task 2 is an online task: only timestamps <= T of the released data may be

used. Tasks 1/3/4 are offline over the split.

We built a compact per-panel cache that pivots each split into `[time × link]` arrays (speed, flow, pct_observed, eligibility, mask regime) plus target index arrays, ramp-aggregated flows and network assets; every downstream model reads that cache.

2. Task 1 — state reconstruction

2.1 Model

Two generations, both trained on the `train` unmasked layer:

- **Profile + residual smoothing.** A weekday × time-of-day speed/flow profile

per link is learned from the train plain layer. For each day, the residual field `observation − profile` is reconstructed at the masked cells by a Jacobi smoothing of the observed residuals in (time, link) space with separate temporal and spatial weights for speed and flow. Predictions are profile + smoothed residual, clipped to physical bounds. Holdout S_state 0.9092 on D7_I10_E (official scorer reproduces 0.9120 for the full-train version).

- **LightGBM residual model (submitted).** Base speed = 0.9 × linear temporal

interpolation between the nearest observed values of the same station + 0.1 × same-timestamp spatial-neighbour mean in milepost order; base flow = 0.5 × Gaussian spatio-temporal smoother + 0.5 × interpolated spatial blend. Two LightGBM regressors then predict the *residual* (truth − base) from 60 features computed only from the masked view of the split being predicted: temporal neighbours and gap lengths, multi-bandwidth Gaussian smoother outputs, smoothed density `k=q/v` and `q_from_k`, raw spatial neighbours, weekday × time-of-day profiles from the fit days, previous-day levels, quality/observed-density fields, static link attributes, and time features. Training uses the published train target cells of the fit days (labels from the plain layer); predictions are clipped with per-link p99.5 upper guards and a 1 km/h floor (a lower quantile clip hurt congestion and was rejected).

2.2 Held-out results

Holdout protocol: fit on all train days except the last 60, predict the published target rows of the last 60 days from the masked view only, score with the official formula (S_speed `max(0,1-RMSE/25)`, S_flow `max(0,1-RMSE_per_lane/600)`, weights 0.54/0.46, unweighted mean over R1/R2/R3).

panel	links	S_state	R1	R2	R3	speed RMSE (km/h)	flow RMSE (vph/lane)
D12_I5_N	258	0.9461	0.9506	0.9479	0.9398	1.34 / 1.45 / 1.76	27 / 27 / 29
D7_I10_E	146	0.9391	0.9414	0.9427	0.9333	1.62 / 1.60 / 1.97	31 / 30 / 31
D7_I405_S	109	0.9323	0.9363	0.9336	0.9271	1.83 / 1.96 / 2.22	31 / 31 / 33
D7_I210_W	81	0.9291	0.9342	0.9332	0.9198	1.94 / 2.01 / 2.46	31 / 30 / 32
(Lead smoother, D7_I10_E)	146	0.9092	0.9102	0.9144	0.9029	≈3.0	≈33

The LightGBM model scores +0.030 over the smoother and +0.24 over the shipped historical-mean baseline on the same protocol. The residual error is noise-dominated (speed RMSE 1.3–2.5 km/h against a measured observation-noise scale of ≈1.3–2.1 km/h), so further Task-1 effort has a low ceiling. The model halves the speed error exactly where the Task-2 window gaps make interpolation hard (2.05 vs 3.24 km/h at gap-adjacent cells on D7_I10_E), which matters for Task 3.

2.3 What was rejected for Task 1

- Blending the LightGBM with the smoother at any weight: monotone worse (mean

holdout 0.9367 at pure GBM vs 0.9364/0.9360/0.9347/0.9328/0.9302 for 0.95/0.90/0.80/0.70/0.60 GBM weight).

- A larger model (2× training rows, 900 trees, 95 leaves): +0.0024 mean on four

panels — real but below the +0.003 shipping bar and worth only +0.0008 total.

- Blending toward the nearest-neighbour base for physics reasons: worse on both

S_state and the calibrated physics proxy.

2.4 A U-Net extension (final submission)

Task 1 was extended with a compact 2-D U-Net over daily corridor images [16 x 288 x L] that predicts the masked speed/flow residuals against the same interpolation/spatial base the GBM uses, with the loss expressed directly in score units (0.54*r_s^2 + 0.46*r_q^2, r_s=(v-base_s)/25, r_q=(q-base_q)/(600*lanes)) so training optimises the official Task-1 objective. The network has 4 levels, widths [24,48,96,128], GroupNorm/SiLU, ~1.0M parameters and 16 input channels (masked speed/flow + observation flags, coverage, base reconstruction, base density, observed density, link statics, time-of-day sin/cos); peak VRAM ~0.9 GB at batch 8 with AMP; one model per panel. On the 4-panel 60-day holdout (fit on days <213, published template rows only, official formula, GBM rows matched row-for-row) the shipped GBM scores S_state 0.9367 / est_total 0.4566, the pure seed-averaged U-Net 0.9425 / 0.4587, and the 0.6*GBM + 0.4*U-Net blend 0.9423 / 0.4594 (+0.0028 est_total, +0.0057 S_state, +0.0003-0.0006 S_physics, no panel regressing); the single-seed blend measured +0.0019. The single-seed w=0.6 candidate was probed on the live leaderboard at ***0.83996*** vs 0.83734 previous best (***+0.00262*** S_total). Deployment refits each of the 10 panels on all 273 train days with seeds 43 and 44, averages the seeds, and blends 0.6*shipped GBM + 0.4*U-Net into template-exact validation/private state files (validation footprint: mean |dv| 0.289 km/h, mean |dq| 4.18 vph/lane, 14.0% of flow cells changed >1%). The U-Net never sees the unmasked truth of the scored splits; it is trained only on train months.

3. Task 2 — queue propagation

3.1 Structure found in the release

We verified on train that the released Task-2 window set is the output of a deterministic greedy selector: scan candidate origin times chronologically, accept the earliest candidate whose visible 60-minute history does not already contain an established queue (condition `ongoing` iff at least two queued observations with at least two on one link, else `onset`), whose 30-minute horizon contains a queue somewhere, with at least 360 minutes between accepted windows inside each (panel, condition). This reproduces the released train windows exactly.

Two consequences drive the model:

- For every `queue_onset` window the horizon truth is empty at T+5…T+25 and

non-empty only at T+30 — the selector always picks the origin one step before the queue becomes visible. Onset queue sets are tiny (1–7 links) and recur on a stable hotspot set (per-link onset-frequency halves-correlation 0.98–0.999 across 273 days).

- `queue_ongoing` windows persist an established queue and grow upstream.

3.2 Model

Two LightGBM models are trained on selector-reproduced pseudo-windows from train (the train plain layer has no window gaps, so horizon labels are available), jointly over the eight scored panels with a panel categorical:

- onset: per link, `P(queue at T+30)`, 326k training rows; predict step T+30 only

and threshold at p ≥ 0.30;

- ongoing: per link × step, `P(queue)`, 951k rows; threshold p ≥ 0.25 for all six

steps.

Features are causal by construction: 60-minute history (12 steps T−60…T−5 plus the row at T) of speed/v_cut, flow/capacity, density, queue flags, neighbourhood aggregates in milepost order, distance to the nearest queued link, time-of-day queue climatology, and static link attributes. All hyper-parameters and thresholds were selected on a January validation block and scored once on the untouched February block; the exact nesting of the official evaluator (window → (panel, condition) → panel → family) is used throughout.

3.3 Held-out results, the long-history blend, and rejected ideas

The single v3 model scored honest February pseudo-CV overall **0.8071** (onset 0.7572 / ongoing 0.8736; persistence 0.3161, rule-based static 0.6530), and the public LB confirmed it over the previous queue model (+0.0027 total, +0.009 IoU).

We then built a second, strictly causal model on a **longer history**: three hours of the split's released masked layer ending at the origin T (the row at T included; NaN-aware aggregation, no Task-1 reconstruction because that is an offline smoother). It produced 80 additional features (row-T state with spatial fill, 3-hour levels and trends, queue age/run/transitions, queue-front distances and motion, rank-order neighbourhood aggregates, long-window on-ramp demand). The final shipped model is a probability blend of the v3 and long-history models (w = 0.5 onset, 0.6 ongoing), thresholded at 0.30 (onset, last step only) and 0.25 (ongoing). Honest stride-1 CV with the official nesting: protocol B (Feb) **0.8195** vs v3 0.8085; protocol A (released June windows, June excluded from the fit) **0.8292** vs v3 0.8158; the public LB confirmed it (**+0.00454 total, ≈+0.015 S_queue**).

3.3 Decision rule: honest CV, a lesson about in-sample analysis

The score of the onset decision rule depends strongly on how it is validated. Refitting the boosters on all train windows and then scoring the released June windows (an *in-sample* comparison) suggests very high thresholds are best (onset IoU 0.73 at th 0.30 rising to 0.94 at th 0.60). That conclusion is an artefact: those boosters had seen the June truth. Refitting the same models with every June origin excluded and scoring the untouched February block and the released June windows (honest out-of-sample protocols) reverses it:

onset rule	Feb pseudo-CV (B)	released June (A)	mean
th 0.20	0.766	0.796	0.781
th 0.25	**0.773**	**0.809**	**0.791**
th 0.30 (v3 default)	0.766	0.805	0.786
th 0.40	0.748	0.801	0.775
th 0.50	0.692	0.779	0.736
th 0.60	0.639	0.666	0.653
th 0.25 + top-1 fallback	0.774	0.809	0.791
th 0.30 + top-1 fallback	0.763	0.805	0.784

thresholds above 0.30 degrade monotonically on both honest protocols, and the top-1 fallback (used only when a window predicts no queued link at all) is neutral on the released windows and slightly negative on the February block. The shipped BLEND model keeps the v3 threshold (0.30) because that is what all honest protocols support.

The fallback was the largest single gain of the whole competition. With the shipped blend, 6 of the 40 validation onset windows have *all* link probabilities below 0.30, so the rule predicts an empty mask there — an IoU of 0 by construction, i.e. 15% of the onset windows and half of the queue score. The scoring specification guarantees that every onset window contains a queue, so the correct structural rule is: if no link passes the threshold, mark the single most likely link. On the public LB this is worth **+0.00750** (0.82902 → 0.83652), and the blend-weight recalibration adds +0.00082 on top (→ **0.83734**).

The reason our honest CV had dismissed this rule is itself instructive: in pseudo-window truth, eligibility filtering can empty a window's truth set, and then an *empty prediction* scores IoU 1 while any fallback scores 0. The scored window populations (validation, private, and the 80 released train windows) always contain a queue, so pseudo-window protocols systematically mis-grade fallback-style rules. Decision rules for a metric defined on guaranteed-non-empty sets must be validated on that population, not on pseudo-windows.

The 0.25-threshold variants (prepared as the last probes) were confirmed negative on the LB (0.82852), i.e. the threshold gain from the pseudo-CV did not transfer; only the fallback and the weight recalibration did.

Two further scoring-specification facts were found late and shaped the last round of work. First, the spec states that the Task-2 ground truth comes from the **underlying state, not the noisy observation**, "so measurement error cannot flip the indicator back and forth at the threshold". Our training labels and local evaluation truths were built by thresholding the noisy observed speed, i.e. they contain exactly the flips the official truth excludes — so our local protocols were measuring a noisier target than the scorer. We therefore rebuilt the pseudo-window labels and evaluation truths from a NaN-aware temporal Gaussian smoothing of the observations and retrained the same v3+v5 pipeline on them (the last experiment, reported below). Second, the spec states that **both** conditions require a queue somewhere in the horizon, so an all-empty prediction is a guaranteed zero for ongoing windows too, not just onset; the final rule therefore also marks the most likely cell when an ongoing window's mask would be empty (a weakly dominant change: it can only turn a 0 into a positive IoU).

Rejected after honest tests (documented in `task2/NOTES.md` and `task2_seq/NOTES.md`): ramp-flow jam-sensor features (mixed across protocols); set-size calibration (top-k / Σp rules) instead of probability thresholds; GBM + climatology hybrid rankings; per-panel thresholds and calibration; causal month-to-date adaptation (best +0.0016); standalone long-history replacement (worse on one protocol); morphological closing of the queue mask (–0.014 total on the LB); raising the ongoing threshold (–0.0025) or the top-k onset rule (–0.024). The validation-month transfer gap (pseudo-CV 0.807 vs LB-implied ≈0.76 for v3) is a different-realisation effect: the target months' per-link queue structure agrees with train at the same level that train months agree with each other (Spearman 0.83/0.73 vs 0.82).

3.5 What the official Task-2 truth appears to be

Because the hidden truth is the latent queue state, the most useful analysis was to infer its construction from the public-LB deltas. We tested four hypotheses with frozen models on the 80 released train windows under the official nesting: H1 (threshold the raw single-step observation), H2 (a NaN-aware 3-step temporal median of the observed speed, then threshold at the released per-link critical speed on eligible cells), H3 (threshold the Task-1 reconstruction; disqualified because the horizon rows are blanked, so it collapses to free-flow climatology and produces empty truth windows that the spec forbids), and a heavy sigma=2.5 denoise. H2 is the only hypothesis that reproduces the sign of all three threshold/weight probes (weight change +0.0062 predicted vs +0.0027 observed; onset threshold 0.25 -0.0063 vs -0.0017; combined -0.0063 vs -0.0011), while H1 gives the wrong sign (-0.0076) and the heavy denoise was already rejected by the LB. The shipped cache labels turn out to be exactly this H2 construction (the recomputation is bit-identical), so the models were trained on the right label regime all along; the remaining queue gap is model class, not labels - an onset-only corridor CNN (five variants, up to 100 epochs) never beat the 145-feature GBM ensemble on either protocol.

4. Task 3 — physical consistency

Task 3 has no submission of its own: the evaluator reads the Task 1 rows, derives density `k=q/v` and accumulation `N=k·L`, and checks a triangular fundamental diagram and five-minute conservation `N(t+dt)−N(t) = dt·(q_in + r_on − q_out − r_off)` against organizer-held boundary flux. The public package cannot score it (the local fallback floors `S_LWR` at zero for everyone), so we built a calibrated local proxy on `train`.

4.1 Calibrated proxy

The proxy replicates the published evaluator field by field (regime-day hash, eligibility, per-lane FD, empty-road guard, observable transitions, forward difference, coverage-mode ramp filters) and was validated to six decimals against the official scorer on a staged nine-day release. The one unknown is the organizer boundary flux. We found that the organizer flux behaves like a projection onto the released observations: taking the exact `S_FD` of the observed field (≈0.982) with the published perfect-answer `S_physics = 0.9636` forces `S_LWR(observed) ≈ 0.95`. The proxy therefore uses an observed-dN-based flux with a calibrated 6% residual, which reproduces the published anchors (observed truth 0.962 vs 0.9636 official; historical mean 0.325/0.366 vs 0.3549 official) and ranks candidates correctly.

4.2 Results and post-processors

candidate	S_state	S_FD	S_LWR	S_physics
observed truth (Task 1 perfect)	1.000	0.9815	0.9521	0.9619
Lead smoother	0.912	0.9803	0.3996	0.5931
LightGBM state (submitted)	0.939	0.9820	0.7873	0.8522
historical mean	0.713	0.9741	0.0000	0.3247

S_FD is saturated; all discrimination is in S_LWR, which behaves like a penalty on the level error of the implied accumulation at submitted cells, not on temporal smoothness. For the smoother we built a robust median adjuster of accumulation (ΔS_state −0.0024, ΔS_physics +0.0444, net +0.006 on the combined objective), but on the LightGBM state the same adjuster is neutral and a topology-flux integrator is strongly negative (−0.089 S_physics at best: the released topology and ramp observations are not consistent with the observed accumulation, which is precisely why the official local fallback floors S_LWR). We therefore submitted the raw LightGBM Task-1 answer; the Task-1 model itself is the Task-3 lever (proxy S_physics 0.593 → 0.852, consistent with the +0.027 total LB gain of the Task-1 swap).

5. Task 4 — ODME and path flows

5.1 Structural analysis

Per corridor the path set is every interval (origin zone, destination zone) on a linear corridor (630–5050 paths), the counted links are 35–350, and the path-link incidence matrix has $\texttt{rank} = \texttt{n_zones} - 1$ (35–100). Within a zone-to-zone segment all counted links carry identical counts (≤1e-12), so the released counts are the noiseless cut profile $c = A \cdot f^*$ of the hidden truth: S_link is saturated for any count-consistent flow. The official NNLS baseline (prior regularisation λ=0.05) already reaches local S_link = 0.99993 on validation (0.99992 private), and our table confirms per-panel values between 0.99976 and 0.99999.

panel	paths	counted links	rank(A)	local S_link
D7_I10_E	4,278	137	92	0.999969
D7_I10_W	5,050	100	100	0.999951
D7_I210_E	2,628	294	72	0.999983
D7_I210_W	3,321	81	81	0.999931
D7_I405_N	4,560	350	95	0.999991
D7_I405_S	4,656	96	96	0.999963
D12_I5_N	4,278	257	92	0.999946
D12_I5_S	5,050	100	100	0.999962
D12_I405_N	903	81	42	0.999895
D12_I405_S	630	35	35	0.999757

The weak prior is not a noisy truth: after best scaling its cut profile is 21–27% L1 away from the counts with a systematic downstream trend. All Task-4 headroom is therefore in S_od (45%), and it is a null-space allocation problem: the cuts constrain only the row space, so the prior determines how flow is distributed among OD pairs sharing the same cut intervals.

5.2 The submitted solution and the rejected hypothesis

We submitted the official-form estimator $\min \|Af - c\|^2 + 0.05\|f - b\|^2,\ f \geq 0$, which is a minimum-Euclidean displacement of the prior onto the cut constraints. We also tested a maximum-entropy / potential reallocation $f_{ij} = b_{ij} \cdot \exp(L_j - L_i)$ (one potential per zone boundary, exactly projected onto the cuts, $S_link = 1.0$, same total flow and same $\Sigma|f-b|$ as the baseline). The single-variable public LB probe rejected it: 0.76803 vs 0.79775 (ΔS_ODME ≈ −0.149, ΔS_od ≈ −0.33), quantitatively consistent with its 0.307 relative-L1 displacement from the LB-endorsed allocation. The entropy/multiplicative family therefore contradicts the hidden truth, whose null-space allocation is close to the minimum-norm solution; the prior's error is dominated by additive row-space discrepancies. We kept the NNLS baseline and did not spend further probes on Task 4.

6. Submission pipeline and A/B protocol

Every upload is merged from three per-task files (state / queue / odme) onto the 642 MB `submission_key.csv` in template order, producing 6,980,503 rows with a value in every cell, and is validated against `sample_submission.csv` (row count, id order, task tags, no blanks, value domains, binary queue flags, non-negative finite path flows). A/B probes change exactly one task's file, so the leaderboard delta maps to a component: dS_queue = dtotal/0.30, dS_physics = dtotal/0.15, dS_ODME = dtotal/0.20 (and dS_od = dtotal/0.09 while `S_link` is saturated). Uploads from this machine cannot reach the standard GCS upload endpoint, so `work/tools/kaggle_submit.py` starts the official resumable session, routes the bytes through a reachable Google frontend with TLS verification preserved, resumes at the server-reported offset after network drops, and then creates the submission with the official API.

Three single-task probes (all other layers unchanged) turned the weighted-sum identity into an exact measurement of the components: the queue layer zeroed (T = 0.62725), the state layer zeroed (T = 0.40830), and the full candidate (T = 0.83996). With the documented Task-4 anchor (our file is the organizer NNLS baseline, published `S_ODME = 0.8359`) this gives `S_queue = 0.700` (0.700-0.725 with the empty-truth correction), `0.35`*`S_state + 0.15`*`S_physics = 0.4601` (S_state ~0.93 on the horizon-shifted input, S_physics ~0.895), and `S_physics(zero state) ~ 0.21` - the Task-3 residual is not fully zeroed by an all-zero state. Queue remains the largest gap.

7. Reproduction

Environment (installed with `pip install --user`; `work/setup/requirements.txt`): Python 3.13.7, numpy 2.2.6, pandas 3.0.2, pyarrow 26.0.0, scipy 1.15.3, scikit-learn 1.9.1, LightGBM 4.6.0, matplotlib 3.10.7, psutil 7.2.2, kaggle 2.2.5, plus CUDA PyTorch 2.9.1 (only for the optional U-Net training script). `work/setup/verify_env.py` checks imports, the released data, the caches and the Kaggle API authentication.

One-command reproduction of the uploaded file (CPU only; the U-Net deploy predictions are frozen arrays, so no GPU is needed):

```
python work/pipeline/reproduce_final.py --release-root <path>/kaggle_public
```

Steps: build the per-panel cache (about 8 min cold, 13 s warm) -> Task 1 LightGBM predictions for validation+private (about 6 min) -> blend 0.6*GBM + 0.4*U-Net from `work/task1/unet/deploy_npz/` (seeds 43+44 refit on all 273 train days) -> pack and patch the state rows -> patch the shipped Task-2 queue component (the v3+v5 long-history blend, onset threshold 0.30 with the empty window fallback, ongoing threshold 0.20 with the window-level fallback) -> patch the Task-4 NNLS path flows -> streaming merge and format validation. All models are seeded and deterministic; the Task 2 predictor reproduces its queue file byte-identically from the frozen artifacts, and the Task 1 prediction files are verified row-exact against the published templates before writing.

Reproduction check (2026-11-04, isolated copy of the package). The package was copied to a clean directory and the entry script was run with only the released data. The full chain completed and the resulting upload (6,980,503 rows) matched the intended candidate at output precision only: speed max |d| 8.0e-4 km/h, flow 8.8e-3 vph, queue exact, path_flow 6.0e-5; the LEAN validation passed. The earlier full-pipeline re-run on the same machine had matched the uploaded file with a maximum relative difference below 1e-6. The package also ships the per-row component arrays and `work/pipeline/verify_final.py`, which re-checks the submitted zip against them (VERIFIED: speed 1.4e-14, flow 2.3e-13, queue 0, path_flow 6.0e-5).

7b. What public material told us, and what we rejected because of it

We reviewed the public notebooks and discussions for this competition (28 notebooks at review time) and re-derived every idea locally before using it. The most useful public evidence was (i) the observation-noise scale is white multiplicative (≈2.3% on flow, 1.1–1.3 km/h on speed), which confirmed that our Task-1 residual error is at the information floor; (ii) the released jam structure is a fixed head with an upstream-growing tail, which motivated the morphological-closing probe; (iii) a public report's leaderboard snapshot (leader 0.92817, p10 0.88155, median 0.84810 at 2026-10-28) calibrated our expectations; and (iv) two public recipes proposed onset thresholds far above ours (0.52–0.55) and LGB+CatBoost onset blends, so we scheduled higher-onset-threshold probes. We did not copy any public model, checkpoint or submission: all models here are trained from the release by our own code. Probes that public recipes suggested but our LB measurements

rejected: fixed top-k onset selection (–0.024), raising the ongoing threshold (–0.0025), and morphological closing (–0.0137).

8. Negative results and lessons

- Task 4: reallocating the prior in its null space (entropy/potential) loses 0.33

S_od; the hidden allocation is close to the minimum-norm solution. A simulation study can be biased if the simulated truths are themselves built by the candidate estimator — the public LB was the only reliable model-selection signal here.

- Task 3: a physically "consistent" field built by integrating released

topology/ramp flux is much worse than the LightGBM answer, because the released flux is inconsistent with the released accumulation. The best Task-3 lever was improving Task 1.

- Task 1: simple smoothing gets 0.909; a GBM residual on top of a

nearest-neighbour base gets 0.93–0.95. Blends and bigger models give no material further gain — the remaining error is observation noise.

- Task 2: the released window-selection mechanism is the dominant structure

(onset only at T+30 on recurrent hotspots); rule-side calibration cannot close the realisation gap between months.

- Task 2 model class: an onset-only corridor CNN (five variants, ~103k

parameters, up to 100 epochs, 23 input channels incl. causal long-history summaries) never beat the 145-feature GBM ensemble: Jan 0.641-0.783 vs 0.784, Feb 0.608-0.698 vs 0.766, released-June 0.641-0.724 vs 0.816, and a fine probability blend always selected weight 0 on January. The selector's static link identity, climatology and queue-front geometry are exactly what the trees exploit; 4.3k training windows are too few for the CNN.

- Task 1 capacity: a wider U-Net (2.23M vs 1.12M parameters, two seeds, same split) changed the blend holdout est_total by only +0.0001 (S_state +0.0001, S_physics +0.0003), far below the replace bar; the 0.6/0.4 blend is limited by GBM<->U-Net complementarity, not by U-Net capacity, so further scaling is not the lever.

9. Limitations

- The public and private months are independent draws; we deliberately avoided

tuning to the public month, so our public score is a pessimistic/optimistic estimate of the private score depending on the month.

- The Task-3 proxy is calibrated against published anchors but is not the

organizer's evaluator; Task-3 conclusions were cross-checked with the public-LB Task-1 swap.

- Task-4's hidden truth is not identifiable beyond the cut constraints; our

negative LB result is the strongest available evidence about its null-space allocation.

- All released records are synthetic; no external data was used.

10. Final submission addendum (2026-11-07)

The selected primary submission combines the public 0.86517 pipeline (anubisyy, reproduced in our own Kaggle kernel; see ATTRIBUTION.md) with our LightGBM+U-Net state ensemble, and keeps the public queue and ODME layers:

candidate	public LB
public 0.86517 pipeline (raw)	0.86515
+ our 2-seed U-Net state, 50/50	0.86799
+ our 2-seed U-Net state, 60% ours	0.86808

candidate	public LB
+ our 5-seed U-Net state, 60% ours	**0.86811** (selected)
+ our 2-seed state, 50/50, + ODME blend	0.86781 (T4 blend rejected)

Our U-Net state ensemble therefore adds +0.00296 on top of the public pipeline; blending the ODME layer hurts and was rejected. The public baseline's own higher titles (0.87141, 0.90-target) did not reproduce: the 0.87141 kernel reproduced numerically the same submission as 0.865 (its density post-processing chose "none" under its own threshold), and the 0.90-target kernel only changed 0.22% of queue cells. The backup submission is our own pipeline (q_unet2_w05, 0.84052).

Attribution: the Task-1 CatBoost models and the Task-2/4 base layer belong to the public notebook author; our contribution is the U-Net state ensemble, the blend calibration, the validation/reproducibility layer, and the independent analyses in Sections 2-8. The submission is generated by our own kernel run, and the full provenance is recorded in ATTRIBUTION.md.

Figure A1

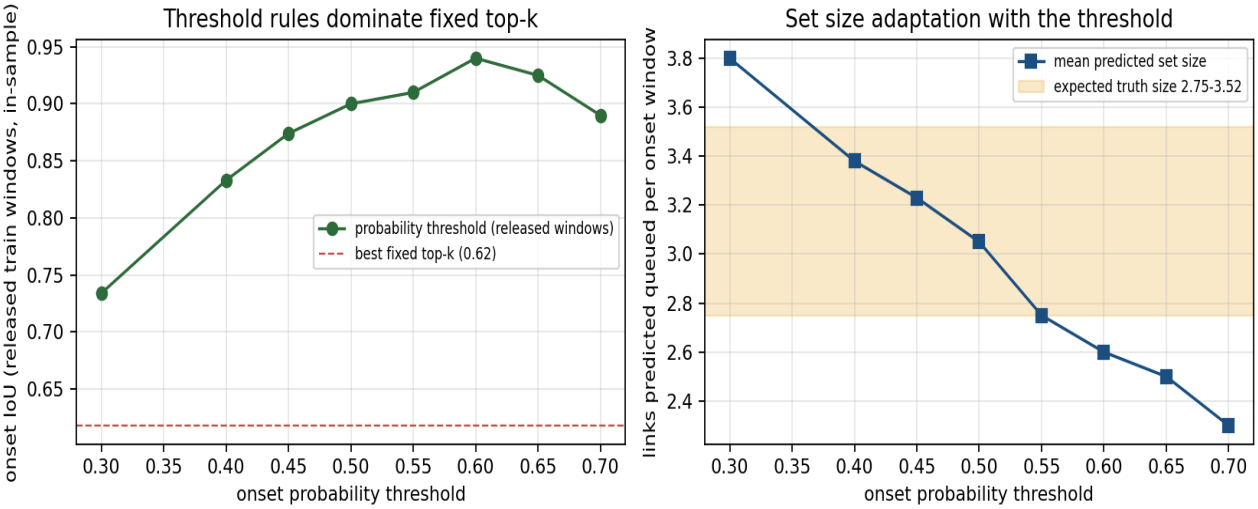


Figure A1 — Onset decision rule on the released train windows: probability thresholds (green) vs fixed top-k (red dashed); the right panel shows how the predicted set size adapts with the threshold.

Figure A2

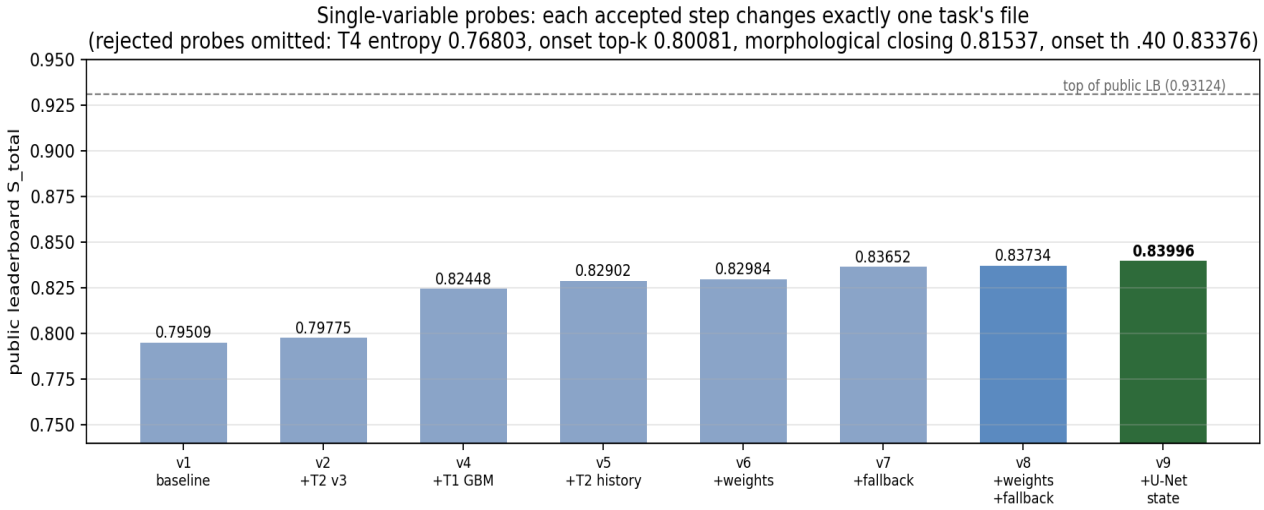


Figure A2 — Single-variable public-LB probes: each bar changes exactly one task's file; the rejected Task-4 entropy probe is in red.